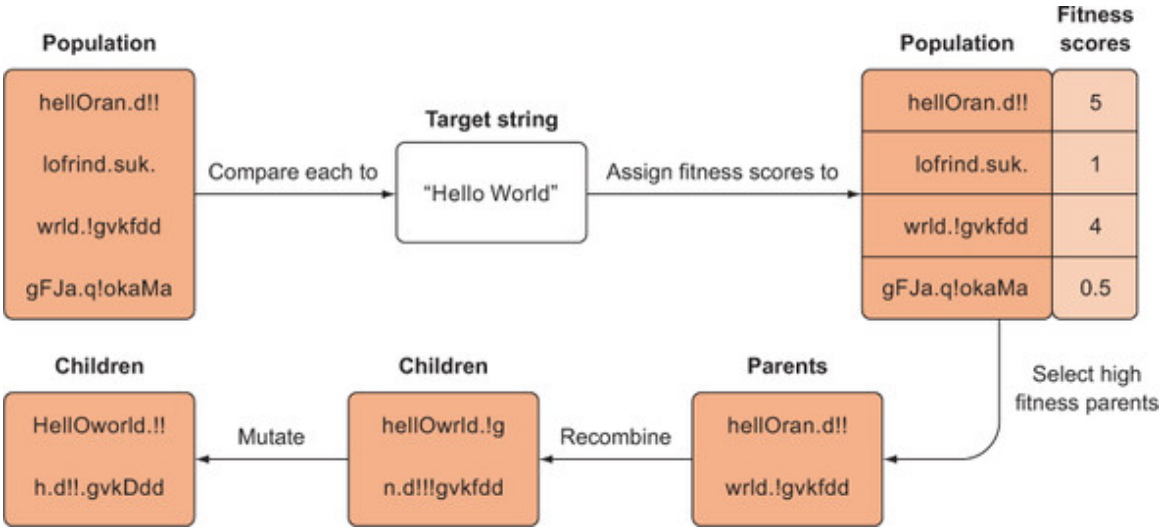


This is perhaps a silly example, but it's one of the simplest demonstrations of a genetic algorithm, and the concepts will directly transfer to our reinforcement learning tasks. Listings 6.1 through 6.4 show the code.

In listing 6.1 we begin by setting up the functions that will instantiate an initial population of random strings and also define a function that can compute a similarity score between two strings, which we will ultimately use as our fitness function.

Listing 6.1. Evolving strings: set up random strings

```
import random
from matplotlib import pyplot as plt

alphabet = "abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ,.!
"   1
target = "Hello World!"
2

class Individual:
3
    def __init__(self, string, fitness=0):
        self.string = string
        self.fitness = fitness

from difflib import SequenceMatcher

def similar(a, b):
4
    return SequenceMatcher(None, a, b).ratio()
```